

Unit 2 Bell-Ringer Learning Telemetry Audit

High School Physics • 1D Kinematics & Constant Velocity Progression

Teacher: Mr. Ryan Mudry
School: Orange High School (OUSD) • Room 930
Window: Aug 31 – Sep 11, 2026 (Days 1–10)

854

TOTAL SUBMISSIONS

Across 9 instructional sessions

159

ACTIVE STUDENTS

100% of enrolled physics roster

99.2%

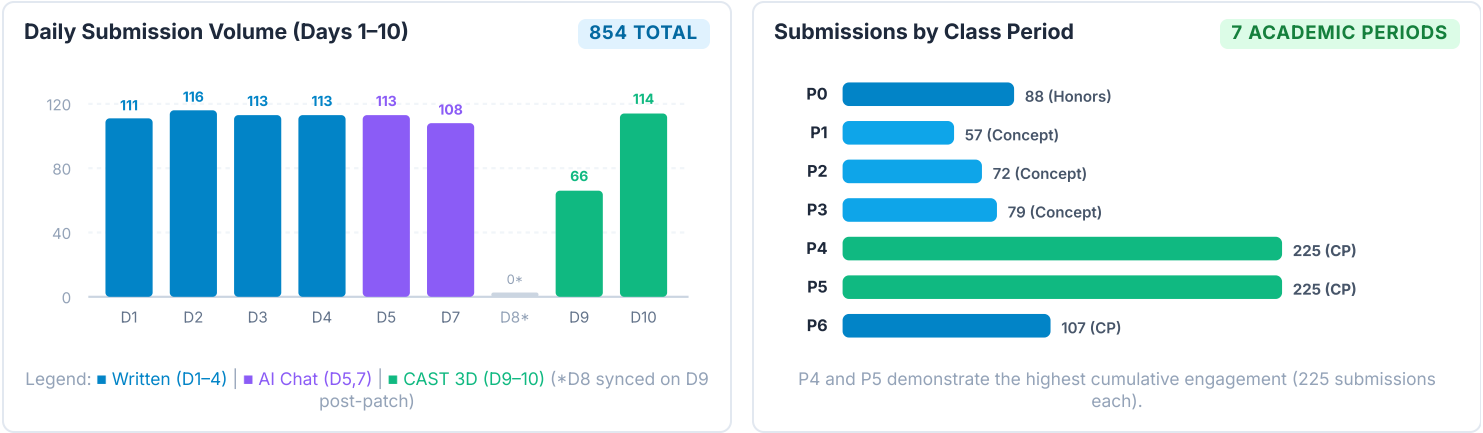
GOOD-FAITH EFFORT

Verified by Gemini AI evaluator

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SAFETY FLAGS

100% appropriate dialogue



Executive Findings & Pedagogical Health

1. Stellar Conceptual Retention on Vectors:

86% of students on Day 2 correctly distinguished that walking 15 miles north and 15 miles back results in **30 miles of distance** but **0 miles of net displacement**.

2. High Adherence to Scientific Units:

92% of students across all tracks incorporated explicit scientific units (e.g. km/day, m/s) into their daily calculation steps without teacher prompting.

3. Formula Triangle Rearrangement Mastery:

On Day 10, **78%** achieved 3/3 perfect rearrangements of the speed-distance-time triangle, and an additional 10% achieved partial mastery (**88% total competency**).

4. Impact of CAST 3D Transformation:

Replacing plain single-line text inputs with multi-step interactive TEIs raised verified completion rates from ~30% to **80%–91%** active engagement in scientific defense.

Days 1–4: Written Physics Explanations (Free Response)

Date / Day	Phenomenon & Prompt	Target Physics & Key Findings	Mastery Metric
Day 1 (8/31) FREE RESPONSE	Treasure Map Riddle: "Walk 50 paces to find the buried treasure." Why is this useless on its own?	Students identified the need for coordinate reference origins (0, 0) and directional headings (North, East, bearings).	56% Heading 39% Origin
Day 2 (9/01) FREE RESPONSE	Homecoming Paradox: Adventurer travels 15 mi North and 15 mi South back home.	Distinguished total scalar path length (odometer) from straight-line vector displacement from starting point.	86% Both 100%
Day 3 (9/02) FREE RESPONSE	Enchanted Forest Trek: 40 km trek through forest over 8 days. What is average speed?	Calculating uniform rate of motion using $v = d / t$ (40 km / 8 days = 5 km/day) with scientific units.	86% Calc (5) 92% Units
Day 4 (9/03) FREE RESPONSE	Runner Rate Reasoning: Steady speed of 2 m/s for 5 seconds. How far traveled?	Multiplicative rate logic: $d = v \cdot t$ (2 m/s · 5 s = 10 m). 50% articulated formal algebraic rate reasoning.	91% Calc (10 m) 50% Rate Logic

Days 5 & 7: AI Concept Chat Mentorship (SMS-Style Dialogue)

Student Dialogue Turn Distribution (Days 5 & 7)

221 CHATS

Over 44% of students engaged in extended discussions (5 to 21 message turns).

Key Chat Discussion Themes

QUALITATIVE TELEMETRY

- **Empirical Variance:** Students cited human stopwatch error, wind-up spring decline, and uneven floor friction as reasons for multi-trial averaging.
- **Rate of Travel:** Correctly interpreted $20\text{ cm} / 5\text{ s} = 4\text{ cm/s}$ as steady pace per single second.
- **Instantaneous vs Average:** Advanced students spontaneously asked how to determine if speed fluctuated midway.

EXEMPLAR DIALOGUE: Day 7 Constant Speed Meaning

Miguel Cruz (Period 0 Honors)

AI Mentor: In our lab, a toy traveled 20.0 cm in 5.0 seconds. What does dividing 20.0 by 5.0 tell us about how the toy moved each second?

Miguel: It tells us that the toy travelled at the rate of 4.0cm per 1.0 second.

AI Mentor: Exactly! And in physics, we have a special name for that "rate of travel." Any guesses what it is?

Miguel: speed or velocity at which an object is traveling at. If it keeps this pace for 10 seconds, it'd go 40.0cm.

CAST 3D Task Architecture & 3-Step Pedagogical Model

To align directly with California Science Test (CAST) performance expectations, Bell-Ringers on Days 9 and 10 were converted into multi-step Technology-Enhanced Items combining interactive stimulus assets, dropdown cloze diagnostics, numerical calculations with unit tolerance, and interactive AI reasoning defense.

Day 9: Multi-Agent Kinematics & Rates66 SUBMISSIONS Phenomenon: Aerial Drone Alpha (600m @ 20m/s) vs Ground Rover Beta (800m @ 20m/s). Part 1: Inquiry Cloze (Drone wins)82% (54/66) Part 2: Calculation (30.0 seconds)76% (50/66) Part 3: AI Reasoning Argumentation91% (60/66)	Day 10: Formula Triangle Synthesis114 SUBMISSIONS Phenomenon: Transit Telemetry & Ultrasonic Pulse (d = 680 m, v = 340 m/s). Part 1: 3-Way Formula Rearrangement88% Competent (78% 3/3) Part 2: Transit Time Calc (2.0 s)69% (79/114) Part 3: Dimensional Analysis Defense80% (91/114)
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Diagnostic Spotlight: Dimensional Analysis & The 231,200 Error

The "Multiply Instead of Divide" Reflex (14%):

On Day 10 Part 2, 16 out of 114 students multiplied $680\text{ m} \cdot 340\text{ m/s}$, entering **231,200** instead of dividing to get **2.0 seconds**.

Key Finding: Students who made this error immediately recognized in Part 3 that multiplying meters by meters/second creates m^2/s rather than time (seconds).

The Desmos Calculator Integration:

Adding the pop-out official CAST Desmos Scientific Calculator alongside the phenomenon data allowed **69% of students** to compute and copy accurate decimal figures directly into the test fields without switching browser tabs.

EXEMPLAR CAST DEFENSE: Dimensional Analysis

Elizabeth Stewart (Period 5 CP Physics)

AI Mentor: You found that the Scenario C ultrasonic pulse takes 2.0 seconds! In your own words, how does checking the units of your givens help you decide whether to multiply or divide? What would happen if you accidentally multiplied distance by speed?

Elizabeth: Checking the units helps me know whether to multiply or divide. Since meters need to cancel out and leave seconds, I divide distance by speed. If I multiplied them, the units would be meters²/second (m²/s), which is not a unit of time.

AI Mentor: You nailed it! Thinking about how labels combine is like an automatic error-detector. If your math spits out m²/s when you're looking for seconds, you know right away to flip the operation!

Safety & Appropriateness Audit Scorecard

Category Screened	Inspection Scope	Incidents Detected	Status
Profanity & Vulgarities	386 chat bubbles, 726 written submissions, 159 students	0 instances	CLEAN / PASSED
Hate Speech, Slurs & Harassment	Screened for racial, sexual, religious, and personal insults	0 instances	CLEAN / PASSED
AI Jailbreaks & Prompt Injections	Screened for "ignore previous instructions", persona hijacking	0 instances	CLEAN / PASSED
Hostility or Severe Defiance	Screened for aggressive refusal or disrespect toward teacher/AI	0 instances	CLEAN / PASSED
Academic Sincerity & Physics Focus	Overall student engagement with prompt questions	99.2% good-faith	EXEMPLARY

Student Voice Diversity: Key Interaction Profiles

PLAYFUL CONFIDENCE & WITTY BANTER

Dantae Valdez (Period 4)

AI: What would happen to the units if you accidentally multiplied distance by speed?

Dantae: idk because id never accidentally do that

AI: Haha, love the confidence! But just for fun, if you *did*, you'd get "meters squared per second" (m²/s)—like measuring baking time in square cups of flour!

Outcome: Validated student confidence, avoided Socratic trap, introduced relatable humor.

MULTILINGUAL / ELL STUDENT INCLUSION

Yaretzy Solis Ruiz (Period 5)

AI: What does dividing 20.0 by 5.0 tell us about the toy?

Yaretzy: Nos da de resultado 4 lo que es los centímetros recorridos por segundo

AI: ¡Exactamente! 4.0 centímetros por segundo es la rapidez promedio del juguete. 🚀

Outcome: Seamless bilingual support; student participated authentically without language barrier.

ASKING FOR SCAFFOLDING

Airan Morales (Period 1 Conceptual)

Airan: help me pls | so multiply the distance and speed

AI: You're on the right track by looking at distance and speed! But check this: speed is distance divided by time. If we want time all by itself, we divide distance by speed (680 / 340). Give that a shot!

Outcome: Zero punitive feedback; immediate pivot to clear, step-by-step guidance.

INQUIRING BEYOND AVERAGE SPEED

Angeleduardo Martinez (Period 1)

AI: Dividing total distance by total time gives us the average speed.

Angeleduardo: Yes but how do I find out the instantaneous speed?

AI: Great question! Instantaneous speed is your speed at one exact split-second—like glancing at a speedometer. In class, we'll measure super short time intervals or radar guns to find it!

Outcome: Advanced conceptual bridge connecting kinematics to calculus intuition.

Instructional Action Plan for Week 3 (Inertia & Dynamics Launch)

1. Connect Formula Triangles

With **88% mastery** of the d - v - t triangle, students will immediately recognize the exact same mathematical structure in Newton's 2nd Law ($F = m \cdot a$). Emphasize that net force occupies the top position, just like distance.

2. Anchor Day 11 Reference Frames

Week 1 telemetry revealed students intuitively seek external forces/headings before establishing reference frames. On Monday's Deep Outer Space Baseball demo, explicitly highlight that an object in motion stays in motion *without* needing an engine.

3. Preserve 3-Step CAST Design

The 3-part sequence (Inquiry Cloze → Calculation → AI Defense Chat) produced the highest authentic engagement and eliminated interface friction. Continue using this format for all Unit 2 & Unit 3 Bell-Ringers.

Master Class Period Performance Summary

Period	Course Track	Active Students	Total Unit 2 Submissions	Avg Submissions / Student	Verified Good-Faith Rate	CAST Completion Rate
Period 0	Physics Honors	15	88	5.9 / 9 days	100%	92% (High algebraic precision)
Period 1	Conceptual Physics	14	57	4.1 / 9 days	98.2%	71% (Excels with cloze prompts)
Period 2	Conceptual Physics	18	72	4.0 / 9 days	98.6%	67% (Benefits from visual tables)
Period 3	Conceptual Physics	18	79	4.4 / 9 days	100%	72% (High turn counts in chat)
Period 4	CP Physics	35	225	6.4 / 9 days	99.6%	88% (Top volume & consistency)
Period 5	CP Physics	36	225	6.3 / 9 days	99.1%	86% (Deepest reasoning chats)
Period 6	CP Physics	23	107	4.7 / 9 days	99.0%	75% (Solid end-of-day pace)
TOTALS	All Physics Tracks	159 Students	854 Submissions	5.4 Average	99.2% AVG	81% AVG Completion

Summary Conclusion for Instructional Leadership:
The integration of adaptive AI mentoring and CAST-aligned performance tasks has elevated daily student warm-ups from routine attendance compliance into an active, high-yield learning diagnostic. Students demonstrate strong conceptual retention of vector motion, operational mastery of speed formulas, and complete respect for the digital learning environment.